



University of Bahrain
College of Information Technology
Department of Information Systems
B.Sc. In Cybersecurity

SENIOR PROJECT
ACADEMIC YEAR 2025-2026-SEMESTER 2
SECURE NATURAL LANGUAGE TO BASH
EXECUTION - SHELLSENTRY

19 May 2026

prepared by:

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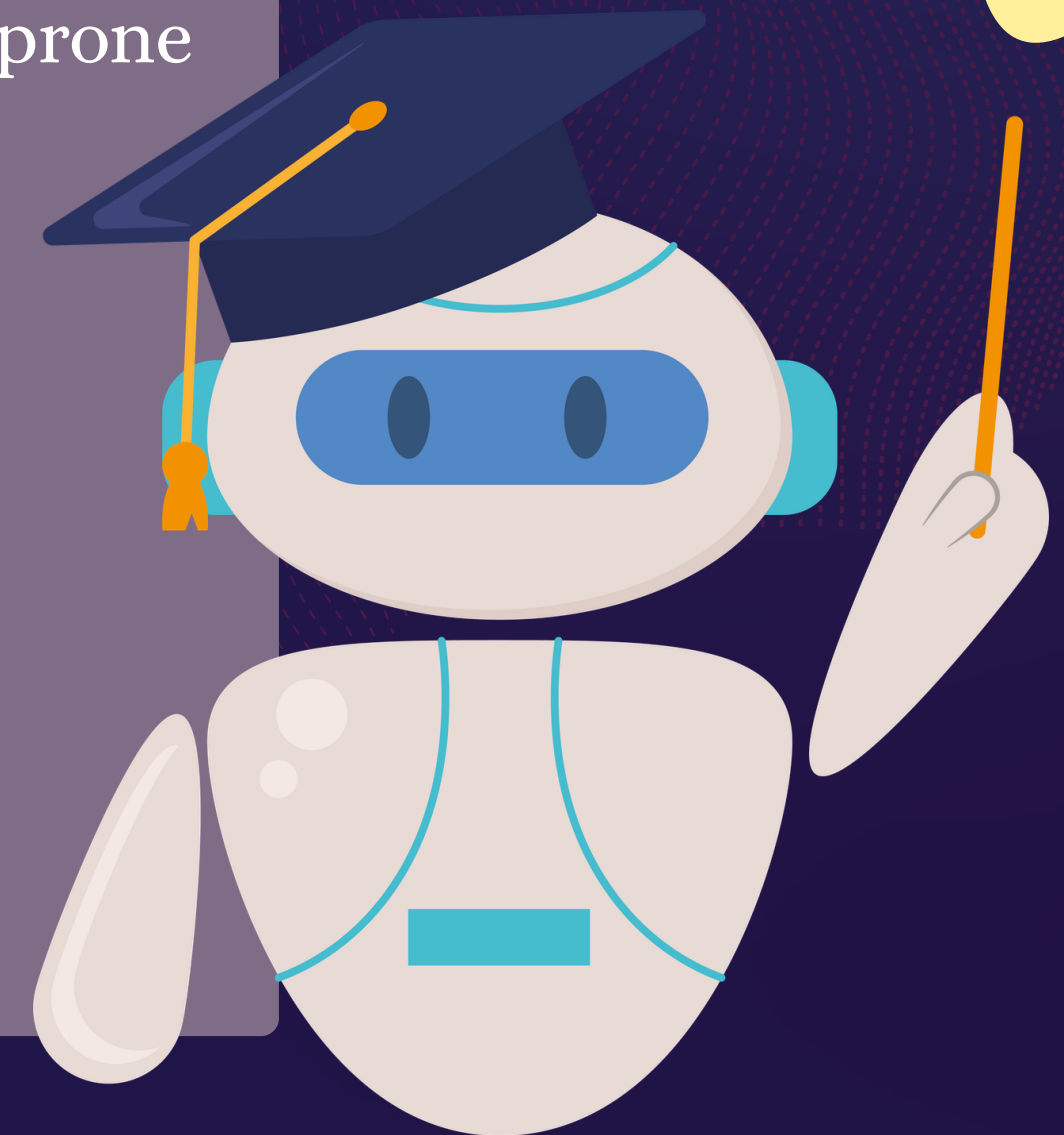
Supervised By :

Dr. Abdulla Khalifa Aldoseri

PROBLEM STATEMENT

- Linux administration requires Bash expertise
- Manual execution across multiple servers is repetitive and error-prone
- Human mistakes can cause outages or security incidents
- LLMs can generate Bash, but outputs may be unsafe:
- Hallucinated destructive commands
- Prompt injection
- Privilege abuse
- Direct remote execution risks

Need for a secure AI-assisted execution platform



OBJECTIVES & CONTRIBUTIONS

Secure NL → Bash
execution

Multi-server SSH
orchestration

Deterministic
command validation

Host-aware command
generation

RAG grounding with
trusted examples

Audit logging and
reporting

Script archive + Safe
scheduling

A major contribution is shifting from “AI assistant” toward a secure operational execution platform.

RELATED WORK SUMMARY

Study	Focus	Limitation
ScriptSmith	Bash generation + refinement	No secure execution
NL→Bash LLM Study	Functional correctness evaluation	No deployment platform
LLM-as-a-Judge	Validation/refinement	Relies on LLM judgment
Shell-GPT	Real-world command assistant	No execution safety enforcement

Research gap: Secure operational execution

RESEARCH GAP

Existing solutions provide:

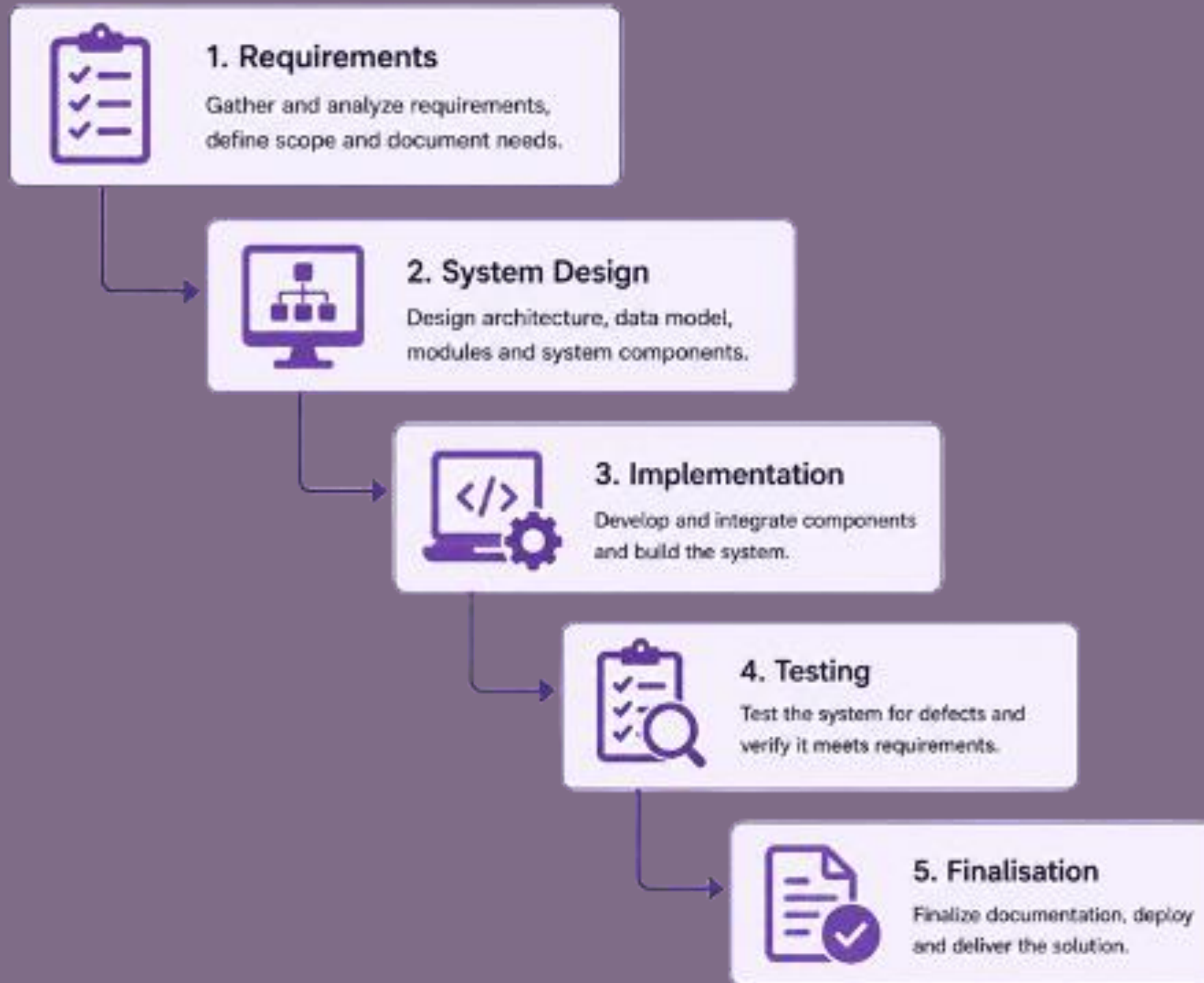
- AI assistance
- Bash generation
- Command refinement

But lack:

- Audit logging
- Safe cron automation
- Deterministic validation
- Authenticated execution
- SSH multi-host orchestration

Then: ShellSentry addresses these gaps

DEVELOPMENT METHODOLOGY



Reason:

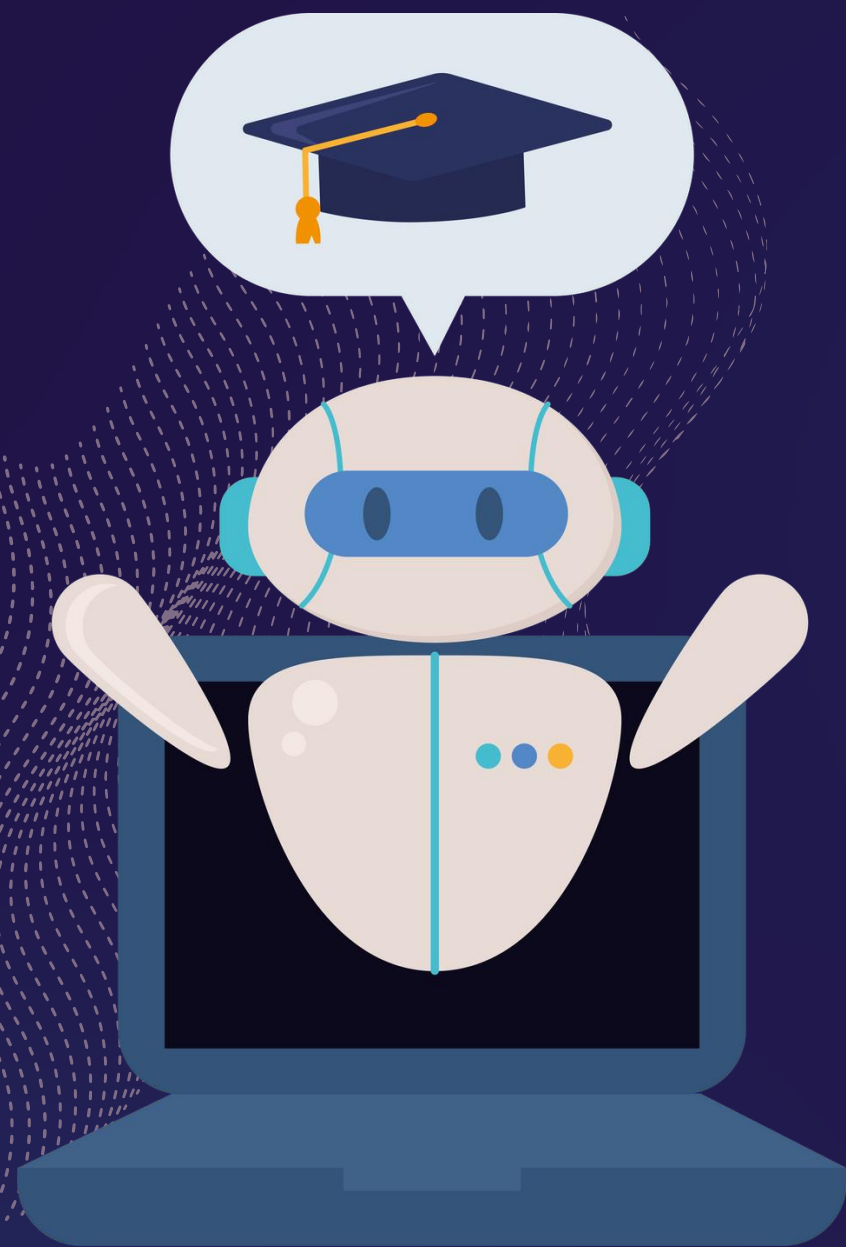
Structured

academic

development

REQUIREMENT COLLECTION

Interviews with 2 cybersecurity specialist



Key findings:

- Read-only by default
- Validation is mandatory
- Auditability required
- Least privilege
- Unsafe LLM output cannot be trusted

FUNCTIONAL REQUIREMENTS

NL → Bash
generation

Host-aware context
collection

RAG retrieval

Command validation

SSH authentication

Multi-server
execution

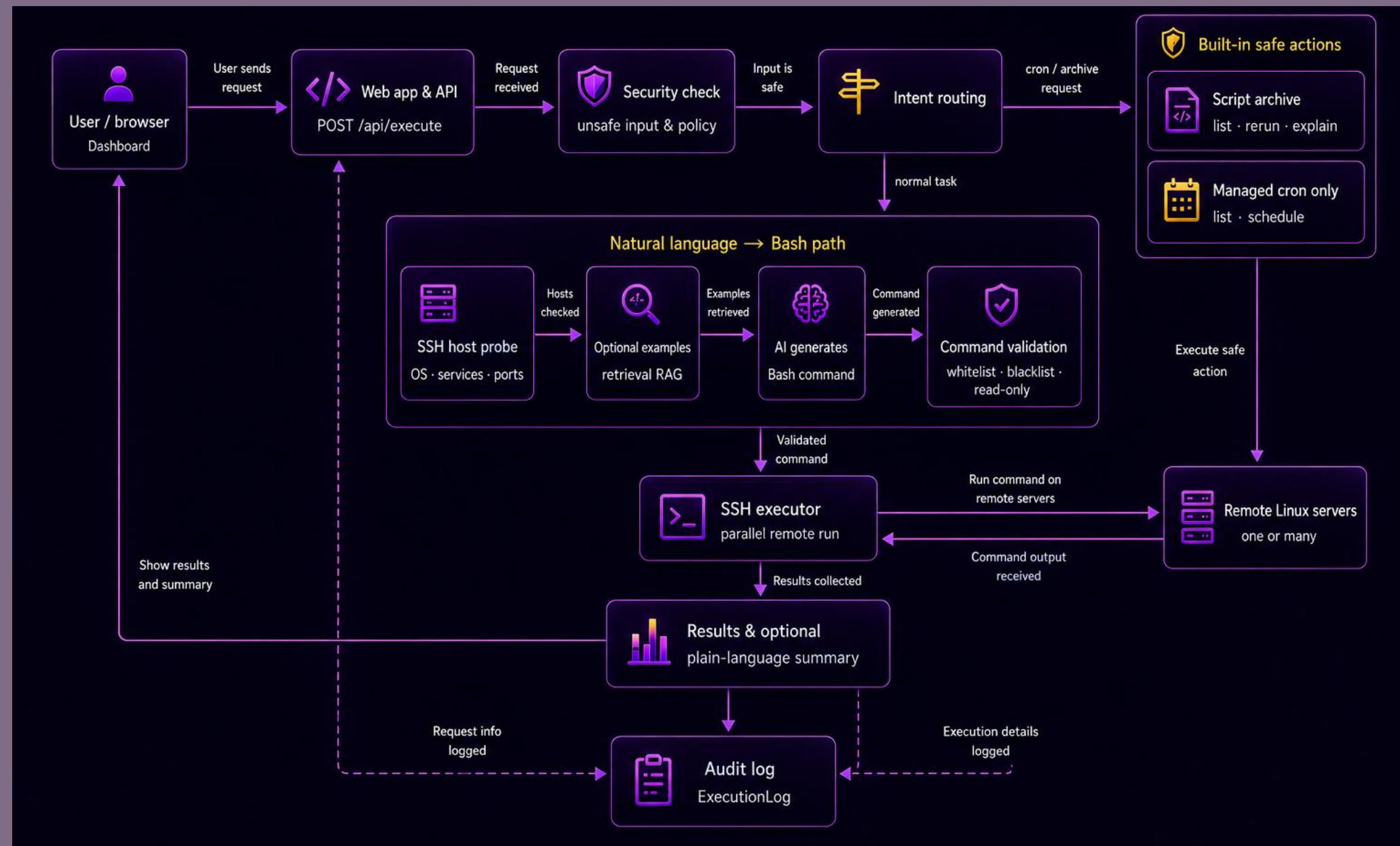
Script archive

Safe scheduling via
Cron

Audit logging

Configurable security
policies

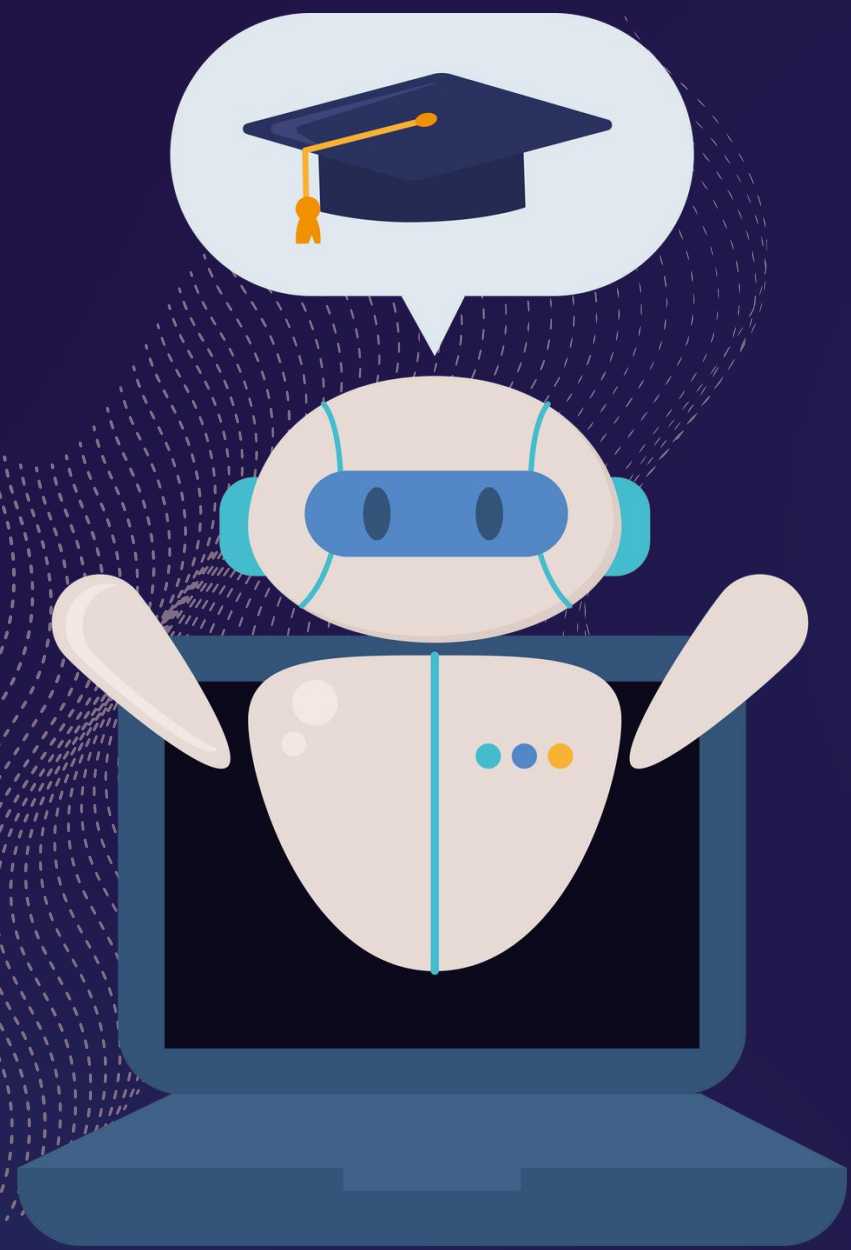
SYSTEM ARCHITECTURE



Explain flow: User → Flask → Security → Routing → RAG →
LLM → Validator → SSH → Results → Database

EXECUTION PROTOCOL





SYSTEM DEMONSTRATION

IMPLEMENTATION STACK

Component	Technology
Frontend	Jinja/HTML/CSS/JS
Backend	Flask
Authentication	Flask-Login
Database	SQLite
SSH	Paramiko
AI model	llama-3.1-8b-instant / llama-3.3-70b versatile
RAG	Sentence Transformers + FAISS

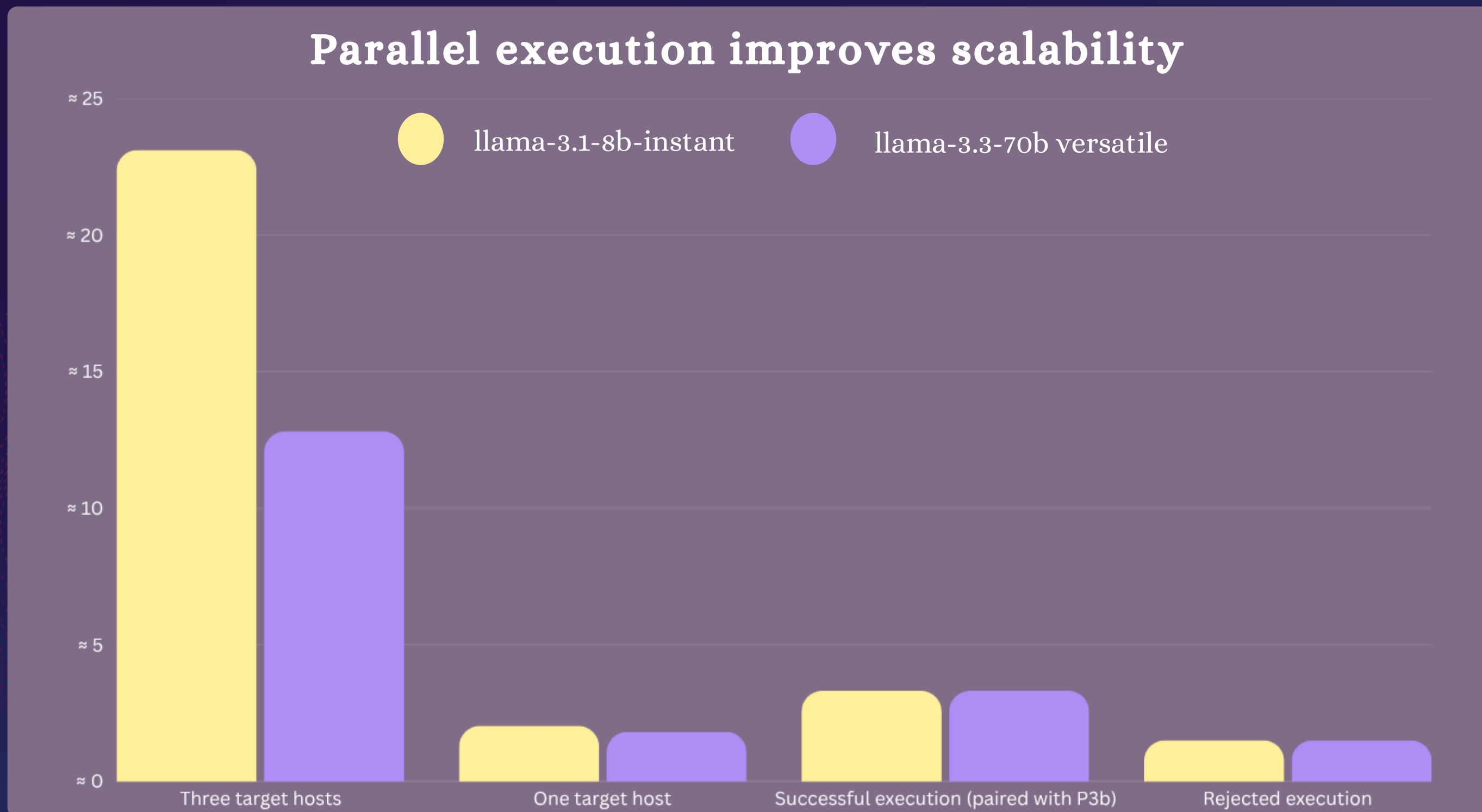
FUNCTIONAL TESTING RESULTS

Test	Result
Authentication	Pass
NL→Bash	Pass
Multi-host execution	Pass
Archive scripts	Pass
Safe scheduling via Cron	Pass

SECURITY TESTING RESULTS

Security Test	Result
SQL Injection	Pass
Prompt Injection	Pass
Unsafe Command	Blocked
Root Execution	Blocked
Cron Deletion	Blocked
SSH Credential Failure	Handled

PERFORMANCE RESULTS



COMPARISON WITH Shell-GPT

ShellSentry focuses on
secure execution, not
just generation

Feature	ShellSentry	Shell-GPT
Bash generation	Yes	Yes
Validation	Yes	Limited
SSH execution	Yes	No
Multi-host	Yes	No
Audit logging	Yes	No
Safe scheduling via Cron	Yes	No



CONCLUSION

- Secure AI-assisted Linux administration prototype
- Validated NL→Bash execution
- Multi-host SSH orchestration
- Strong security enforcement
- Successful testing results

FUTURE Work

- RBAC
- Human approval workflow
- Dry-run mode
- SIEM integration
- Production deployment



THANK YOU FOR YOUR ATTENTION

We sincerely appreciate your time, interest, and support throughout our presentation.

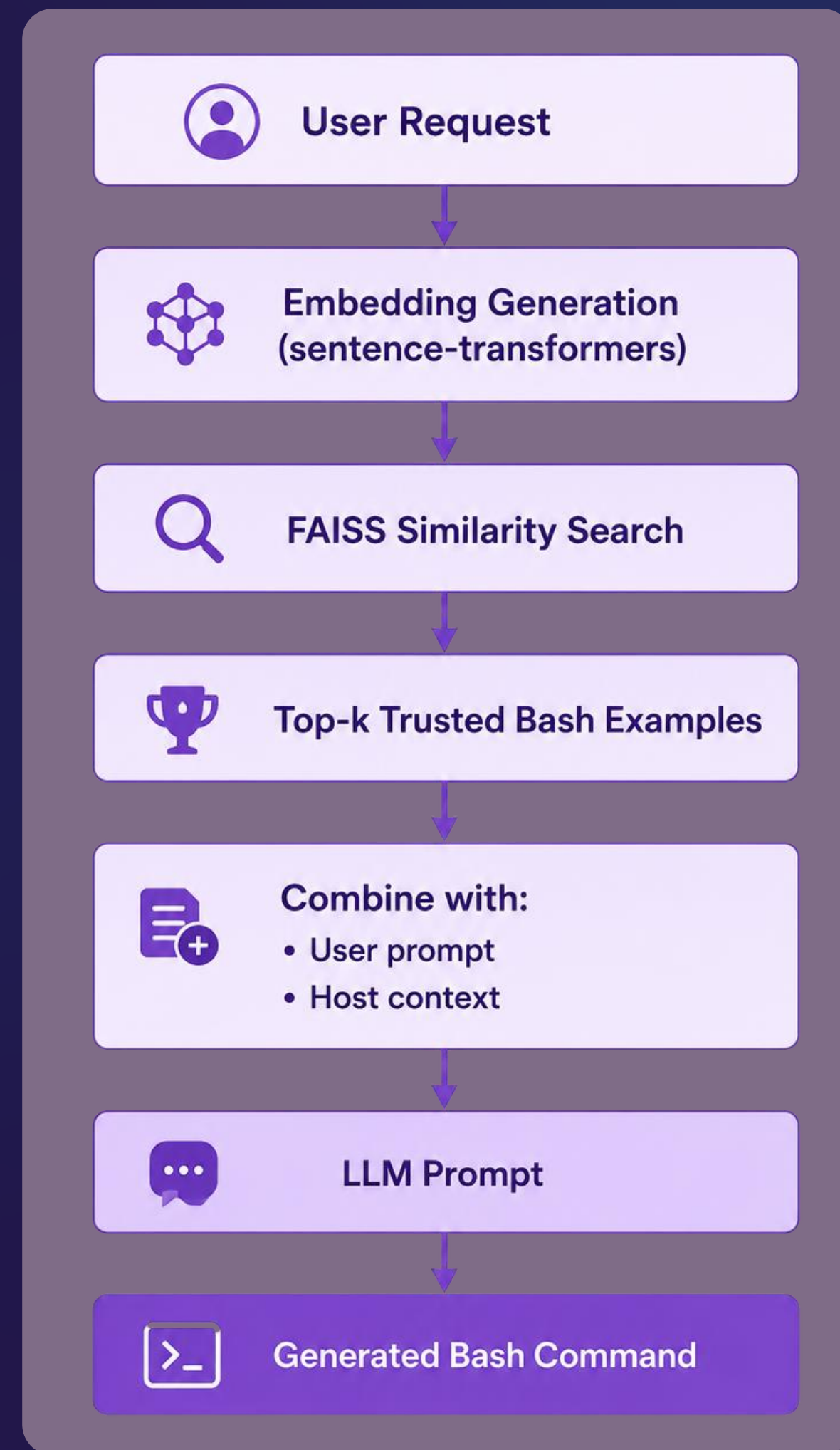
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اللهم لك الحمد حتى ترضى ولك الحمد إذا رضيت ولك الحمد بعد الرضى.

**ANY
QUESTIONS?**

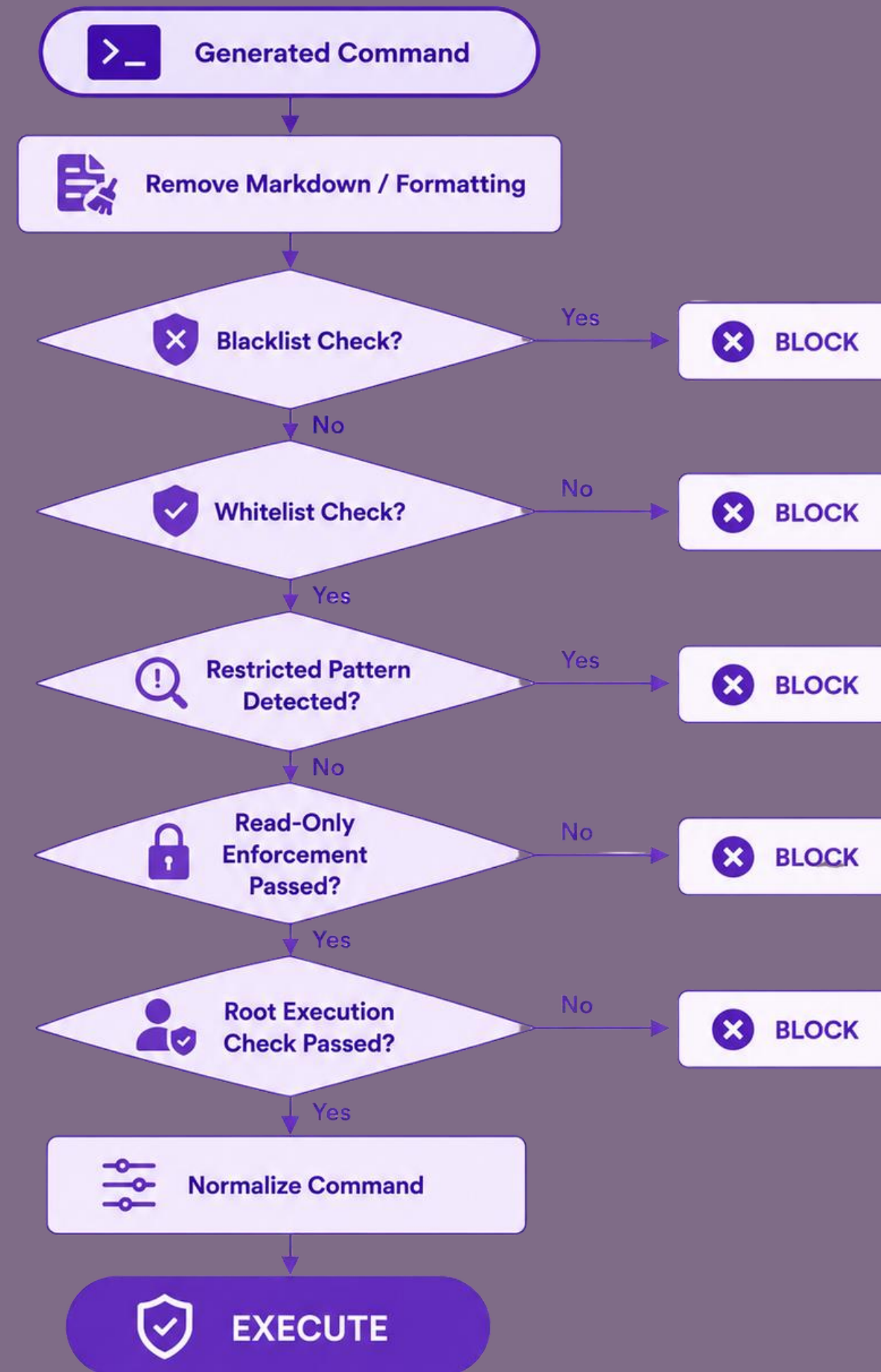


RETRIEVAL AUGMENTED GENERATION PIPELINE



Extras

DETERMINISTIC COMMAND VALIDATION



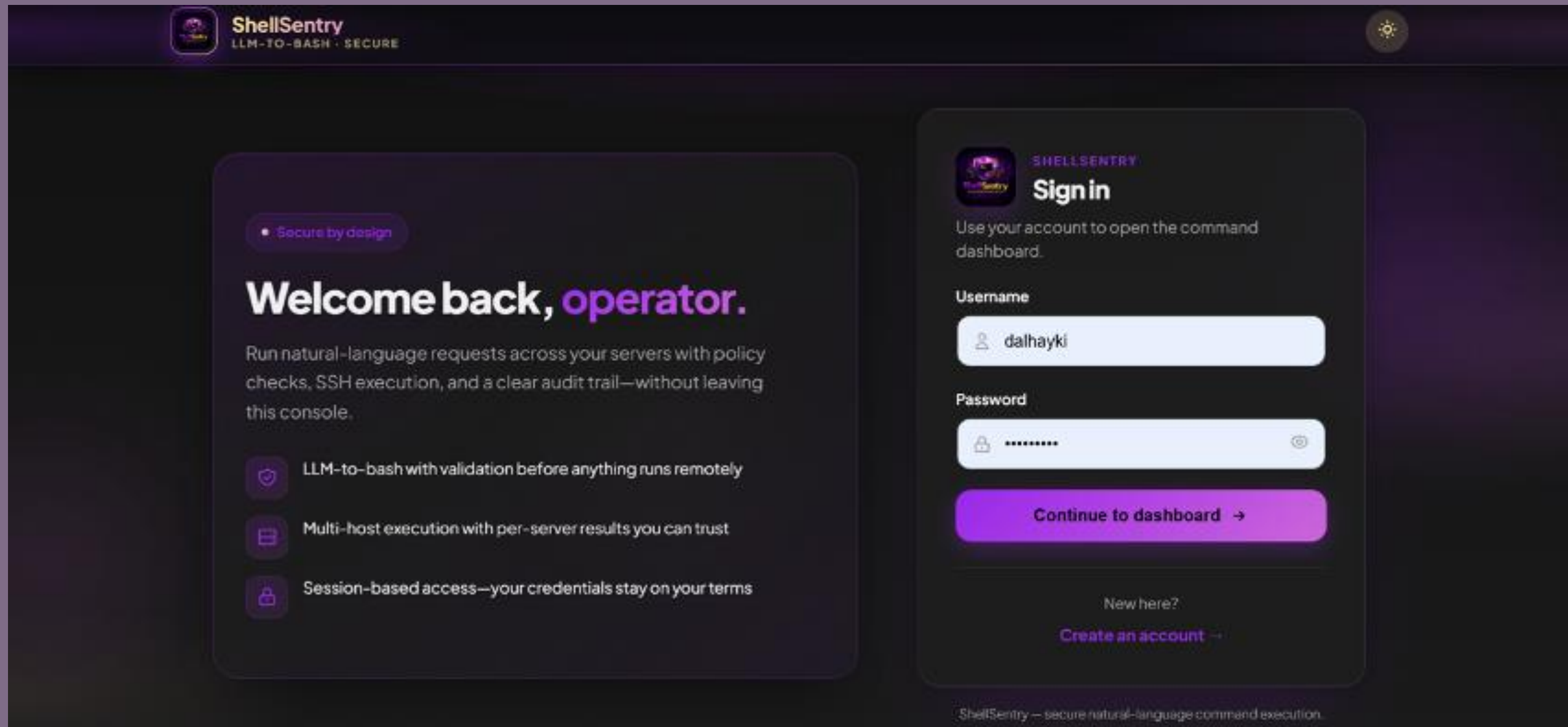
SYSTEM CONFIGURATION CONTROLS

- environment variables
- no hardcoded secrets
- least privilege

Setting	Default
READ_ONLY_EXECUTION	TRUE
ALLOW_ROOT_EXECUTION	FALSE
SSH authentication	Password / Key / Agent
REMOTE_SERVERS	Configurable
SERVER_CREDENTIALS	Per-host supported
LLM provider	Configurable

Extras

SYSTEM INTERFACE



The screenshot shows the ShellSentry login interface. The header includes the ShellSentry logo and tagline 'LLM-TO-BASH · SECURE'. The main content is divided into two sections. The left section, titled 'Welcome back, operator.', features a 'Secure by design' badge and a list of three security features: 'LLM-to-bash with validation before anything runs remotely', 'Multi-host execution with per-server results you can trust', and 'Session-based access—your credentials stay on your terms'. The right section, titled 'Sign in', prompts the user to use their account to open the command dashboard. It contains input fields for 'Username' (with the value 'dalhayki') and 'Password' (masked with dots), a 'Continue to dashboard' button, and a link to 'Create an account' for new users. The footer states 'ShellSentry — secure natural-language command execution.'

ShellSentry
LLM-TO-BASH · SECURE

Secure by design

Welcome back, operator.

Run natural-language requests across your servers with policy checks, SSH execution, and a clear audit trail—without leaving this console.

- LLM-to-bash with validation before anything runs remotely
- Multi-host execution with per-server results you can trust
- Session-based access—your credentials stay on your terms

Sign in

Use your account to open the command dashboard.

Username

dalhayki

Password

Continue to dashboard →

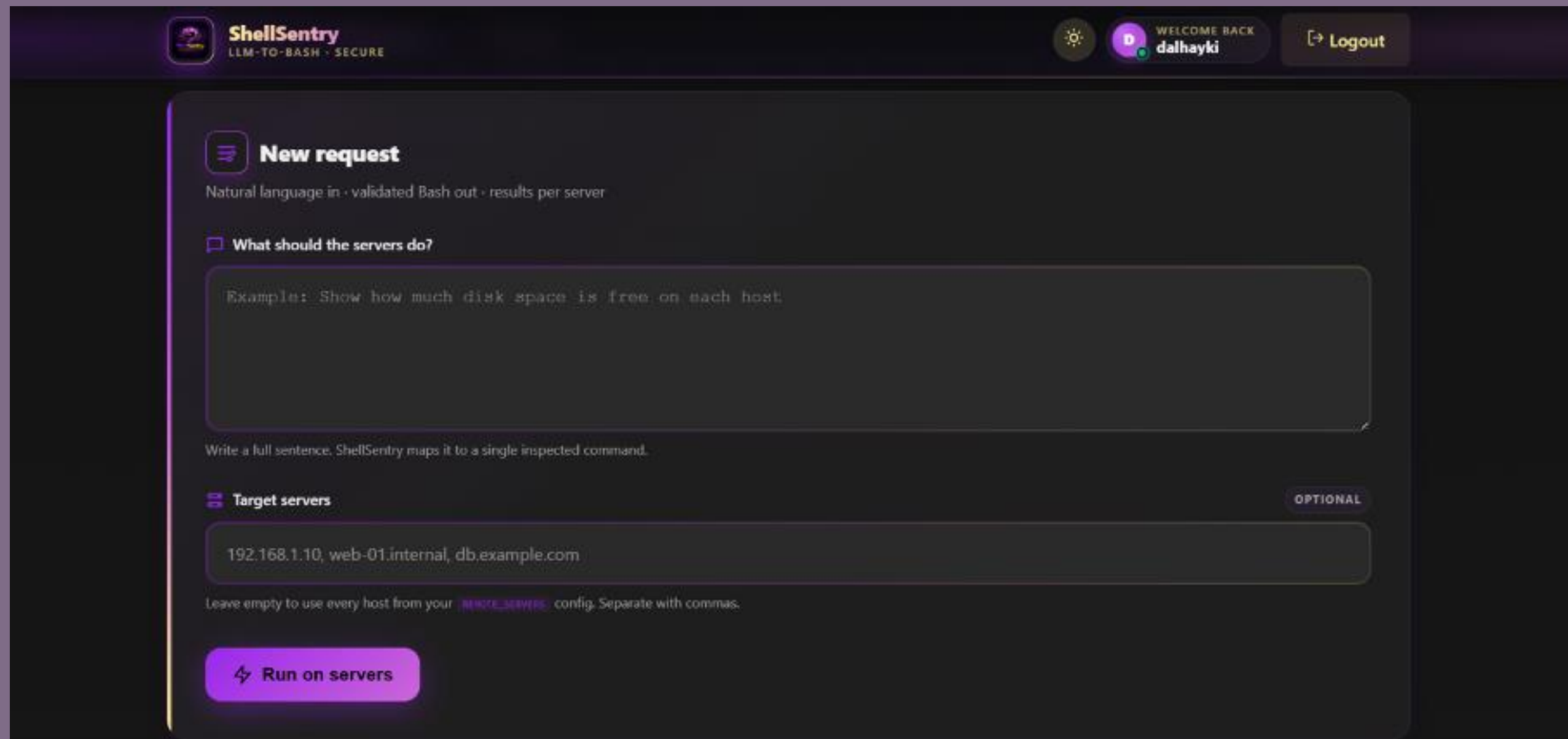
New here?
Create an account →

ShellSentry — secure natural-language command execution.

Login page

Extras

SYSTEM INTERFACE



The screenshot shows the ShellSentry dashboard. At the top, there's a header with the ShellSentry logo (LLM-TO-BASH · SECURE), a settings icon, a user profile for 'dalhayki' with 'WELCOME BACK', and a 'Logout' button. The main content area is titled 'New request' and includes a sub-header 'Natural language in · validated Bash out · results per server'. Below this is a section 'What should the servers do?' with a text input field containing the example: 'Show how much disk space is free on each host'. A note below the input says 'Write a full sentence. ShellSentry maps it to a single inspected command.' The next section is 'Target servers' with a text input field containing '192.168.1.10, web-01.internal, db.example.com'. A note below this input says 'Leave empty to use every host from your REMOTE_SERVERS config. Separate with commas.' At the bottom of the form is a purple button labeled 'Run on servers'.

ShellSentry
LLM-TO-BASH · SECURE

WELCOME BACK
dalhayki

Logout

New request

Natural language in · validated Bash out · results per server

What should the servers do?

Example: Show how much disk space is free on each host

Write a full sentence. ShellSentry maps it to a single inspected command.

Target servers

192.168.1.10, web-01.internal, db.example.com

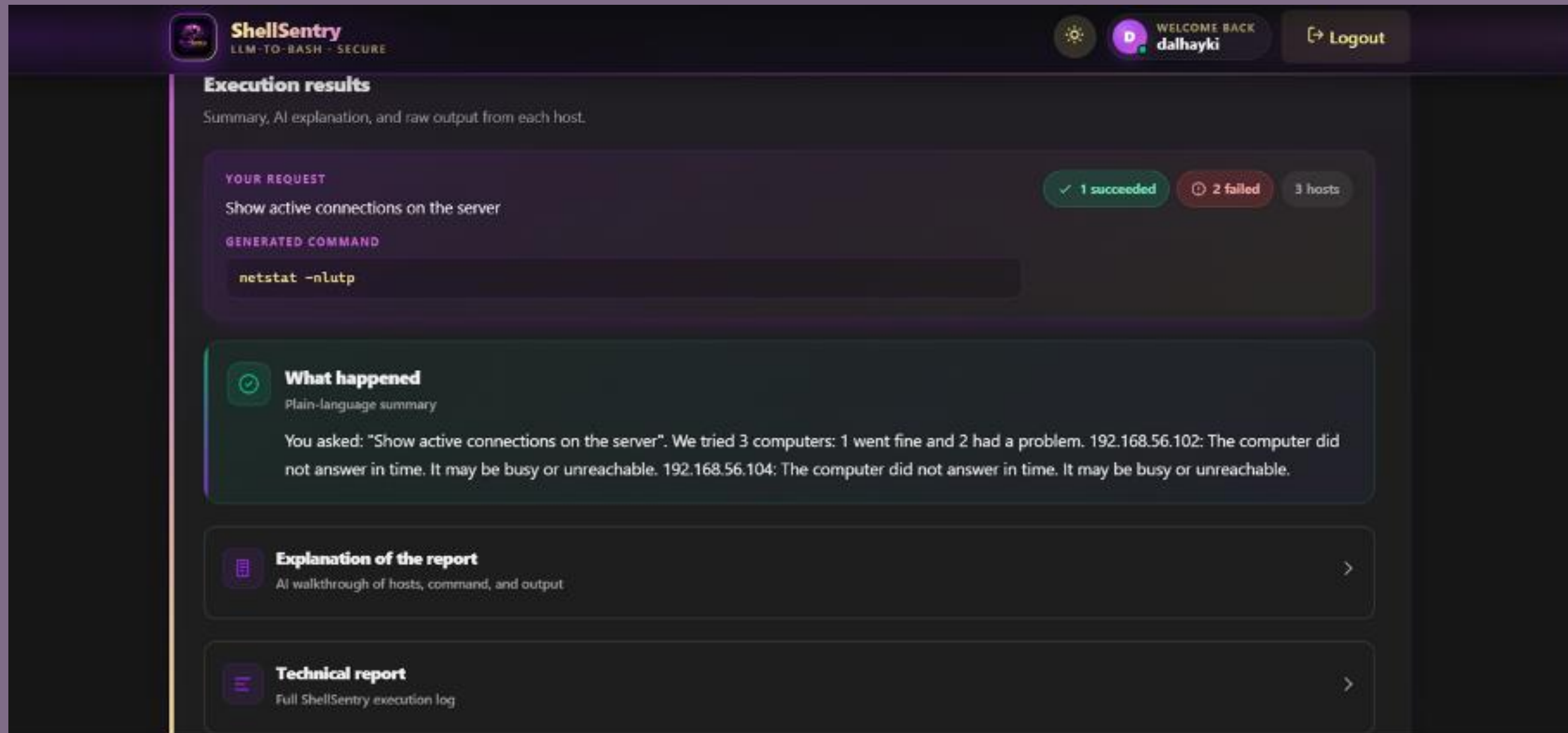
Leave empty to use every host from your REMOTE_SERVERS config. Separate with commas.

Run on servers

Dashboard

Extras

SYSTEM INTERFACE



The screenshot displays the ShellSentry web interface. At the top, the logo 'ShellSentry' is accompanied by the tagline 'LLM-TO-BASH - SECURE'. On the right, a user profile for 'dalhayki' is shown with a 'Logout' button. The main content area is titled 'Execution results' and includes a sub-header 'Summary, AI explanation, and raw output from each host.' Below this, a section labeled 'YOUR REQUEST' shows the command 'Show active connections on the server'. To the right of this section, status indicators show '1 succeeded' (green), '2 failed' (red), and '3 hosts' (grey). Underneath, the 'GENERATED COMMAND' is displayed as 'netstat -nltup'. The interface then presents three expandable sections: 'What happened' (a plain-language summary of the execution attempt on three computers), 'Explanation of the report' (an AI walkthrough of the hosts, command, and output), and 'Technical report' (the full ShellSentry execution log).

ShellSentry
LLM-TO-BASH - SECURE

WELCOME BACK
dalhayki
Logout

Execution results

Summary, AI explanation, and raw output from each host.

YOUR REQUEST
Show active connections on the server

GENERATED COMMAND
`netstat -nltup`

✓ 1 succeeded 2 failed 3 hosts

What happened
Plain-language summary

You asked: "Show active connections on the server". We tried 3 computers: 1 went fine and 2 had a problem. 192.168.56.102: The computer did not answer in time. It may be busy or unreachable. 192.168.56.104: The computer did not answer in time. It may be busy or unreachable.

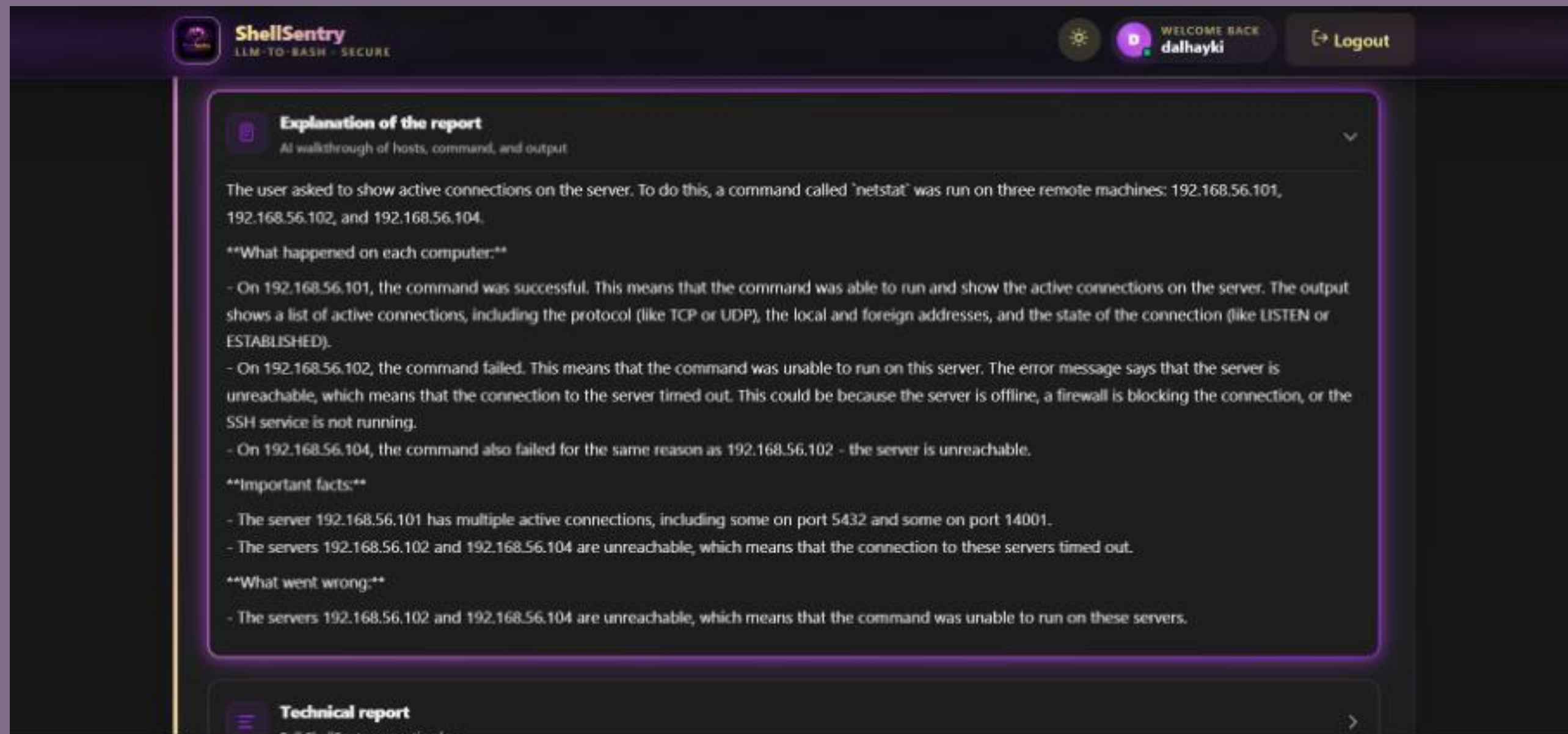
Explanation of the report
AI walkthrough of hosts, command, and output

Technical report
Full ShellSentry execution log

Result page

Extras

SYSTEM INTERFACE



The screenshot displays the ShellSentry web interface. At the top, the logo 'ShellSentry' is accompanied by the tagline 'LLM-TO-BASH - SECURE'. On the right, a user profile for 'dalhayki' is shown with a 'Logout' button. The main content area features a section titled 'Explanation of the report' with a sub-header 'At walkthrough of hosts, command, and output'. The text explains that a 'netstat' command was run on three remote machines (192.168.56.101, 192.168.56.102, and 192.168.56.104). It details the results: the command was successful on 192.168.56.101, showing active connections, but failed on the other two due to the servers being unreachable. A section titled 'Important facts' summarizes that 192.168.56.101 has active connections on ports 5432 and 14001, while the other two servers are unreachable. Finally, a 'What went wrong' section states that the command failed on the unreachable servers. Below this, a 'Technical report' section is partially visible.

ShellSentry
LLM-TO-BASH - SECURE

WELCOME BACK
dalhayki Logout

Explanation of the report

At walkthrough of hosts, command, and output

The user asked to show active connections on the server. To do this, a command called 'netstat' was run on three remote machines: 192.168.56.101, 192.168.56.102, and 192.168.56.104.

What happened on each computer:

- On 192.168.56.101, the command was successful. This means that the command was able to run and show the active connections on the server. The output shows a list of active connections, including the protocol (like TCP or UDP), the local and foreign addresses, and the state of the connection (like LISTEN or ESTABLISHED).
- On 192.168.56.102, the command failed. This means that the command was unable to run on this server. The error message says that the server is unreachable, which means that the connection to the server timed out. This could be because the server is offline, a firewall is blocking the connection, or the SSH service is not running.
- On 192.168.56.104, the command also failed for the same reason as 192.168.56.102 - the server is unreachable.

Important facts:

- The server 192.168.56.101 has multiple active connections, including some on port 5432 and some on port 14001.
- The servers 192.168.56.102 and 192.168.56.104 are unreachable, which means that the connection to these servers timed out.

What went wrong:

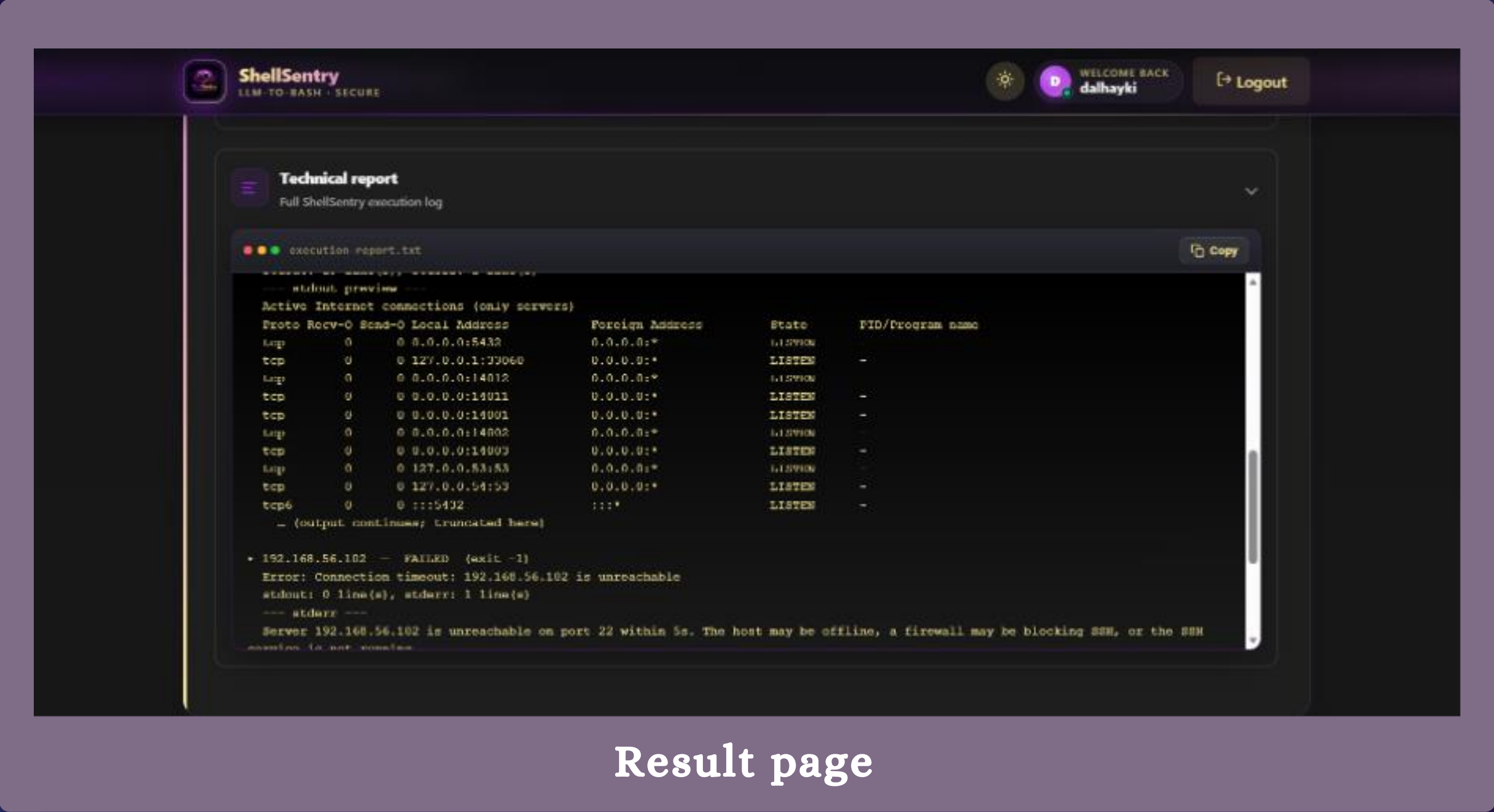
- The servers 192.168.56.102 and 192.168.56.104 are unreachable, which means that the command was unable to run on these servers.

Technical report

Result page

Extras

SYSTEM INTERFACE



Result page